SHELF & SOUL

Technical Project Report | Updated for Supabase PostgreSQL | 19 October 2026

Project and database

Shelf & Soul is a Next.js 15 / React 19 / TypeScript library-management application. Its browser prototype uses Local Storage. Production persists accounts and library state through Prisma to PostgreSQL hosted by Supabase. Supabase is database-only in this integration; application authentication remains custom (bcrypt hashes and signed HTTP-only sessions), with no Supabase Auth or Storage SDK.

Connection setup

- Set DATABASE_URL to Supabase Transaction Pooler (port 6543, pgbouncer=true) for Prisma runtime connections; set DIRECT_URL to a Supabase direct connection or session-pooler URL for migrations and pg_dump.
- Keep database URLs and AUTH_SECRET server-side in an ignored .env file. Never expose them using NEXT_PUBLIC_ variables.
- Run npm run db:generate, npm run db:migrate, and npm run db:seed after configuring a new Supabase project.

Implemented workflows

- Role-aware staff/customer workflows for catalog, circulation, reservations, inventory, members, suppliers, purchasing, audit history and reports.
- Cash-on-pickup orders; no online payment processor. Orders remain unpaid until staff confirms cash collection.
- Password reset uses hashed, single-use, expiring tokens; production email requires provider configuration.
- Encrypted AES-256-GCM PostgreSQL dump script with retention cleanup; scheduling and restore testing remain operator responsibilities.

Persistence and security notes

Most library domain data is currently stored in a JSON LibrarySnapshot row rather than normalized tables. The Prisma server connection uses database credentials; the application relies on its server-side role checks, not Supabase Row Level Security. Protect connection strings and preserve authorization checks.

Validation

- Passed: Prisma Client generation, Prisma schema validation, TypeScript typecheck, and optimized production build with placeholder PostgreSQL URLs.
- The project has not been connected to a real Supabase instance in this workspace. Live migration, database queries, email delivery, and backup/restore were not verified.

Remaining before production

- Create/configure the Supabase project and put its actual connection URLs in local/deployment secrets; apply migrations and seed intended accounts.
- Test production flows against Supabase. Existing MySQL databases and prototype Local Storage data are not automatically migrated.
- Configure email, off-site backup scheduling, and a tested PostgreSQL recovery process. Email verification and normalized relational library tables remain incomplete.

Conclusion: the code is configured for Supabase PostgreSQL and build checks pass, but a real project connection and production readiness still require operator configuration and live testing.